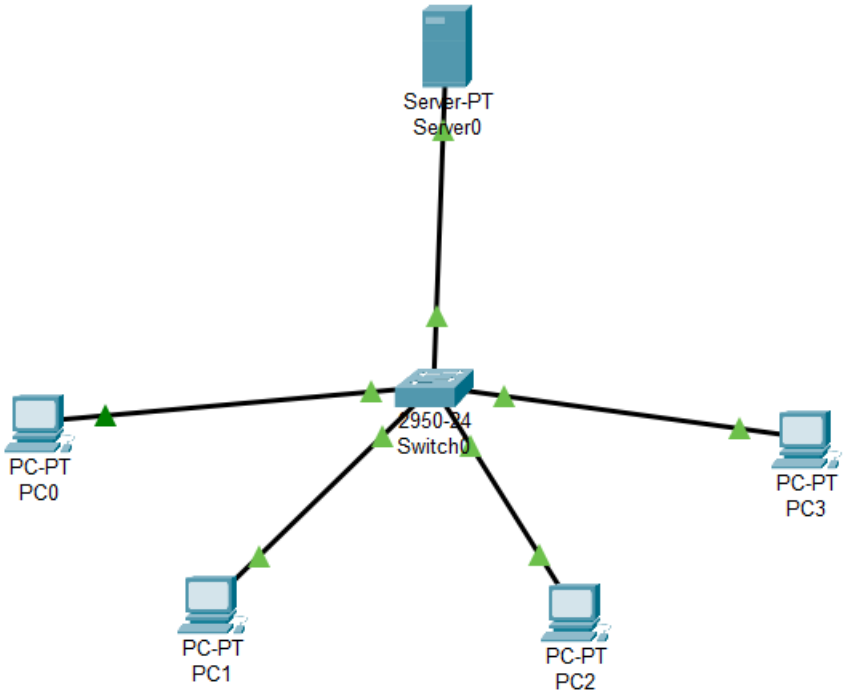
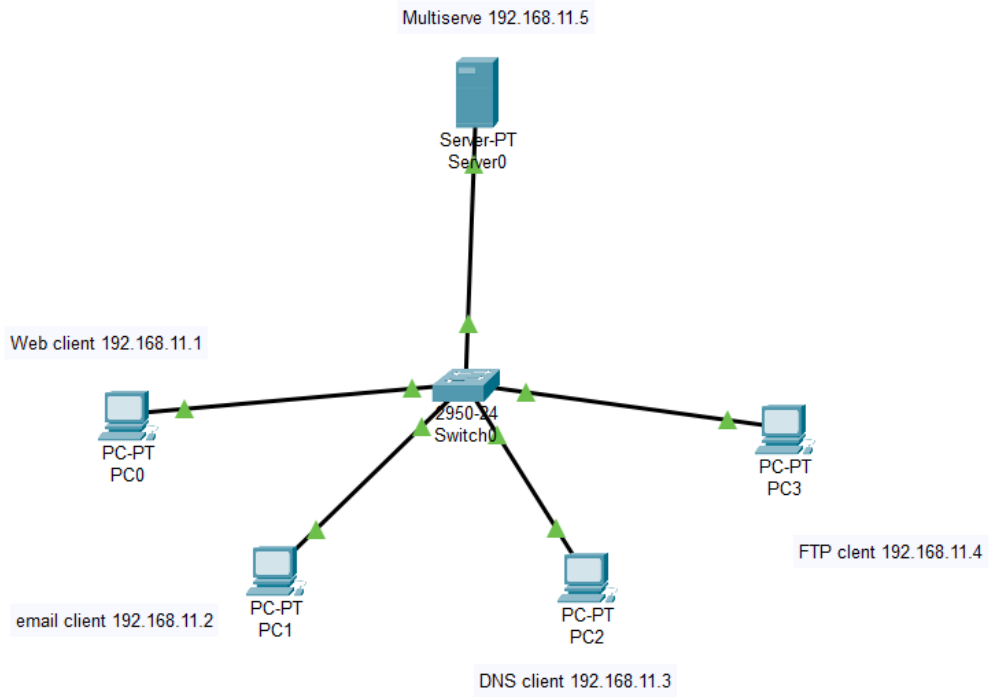
Open Cisco packet tracer

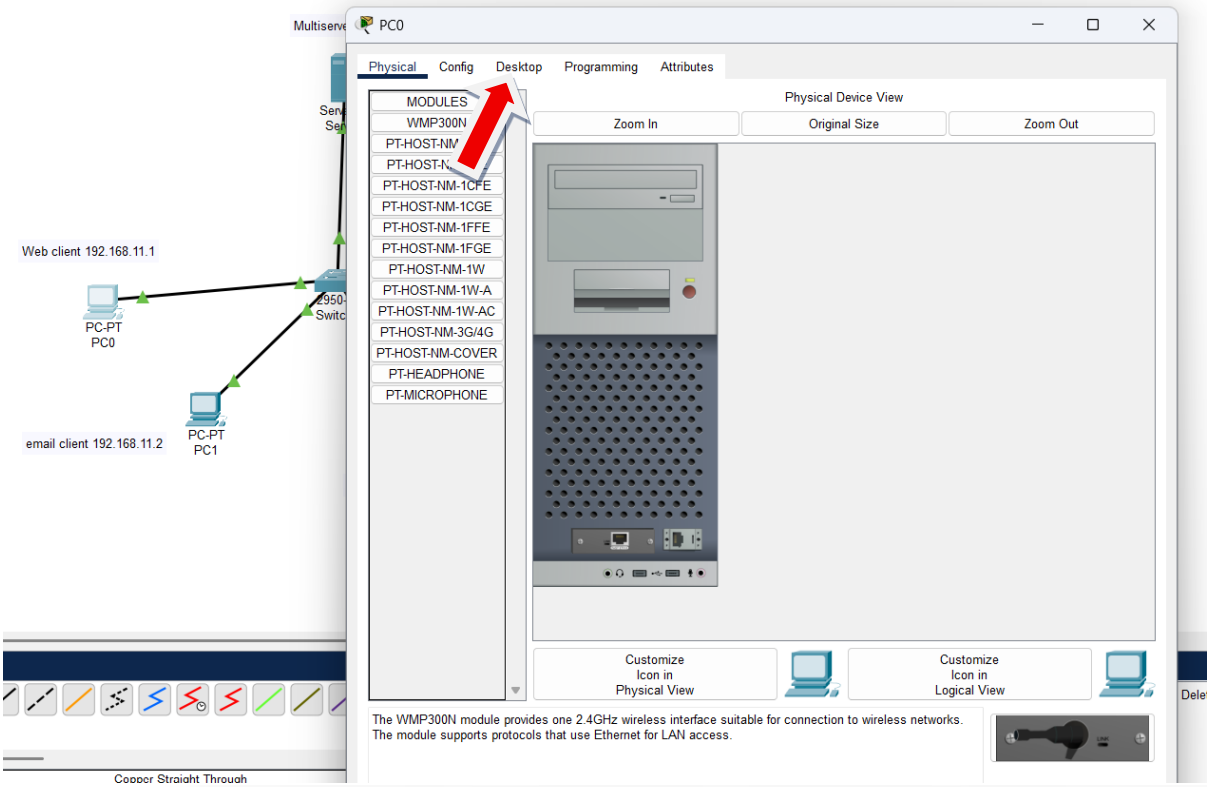
1. Place client nodes and server node with switch and make connection by using copper straight wire ethernet as shown in fig.

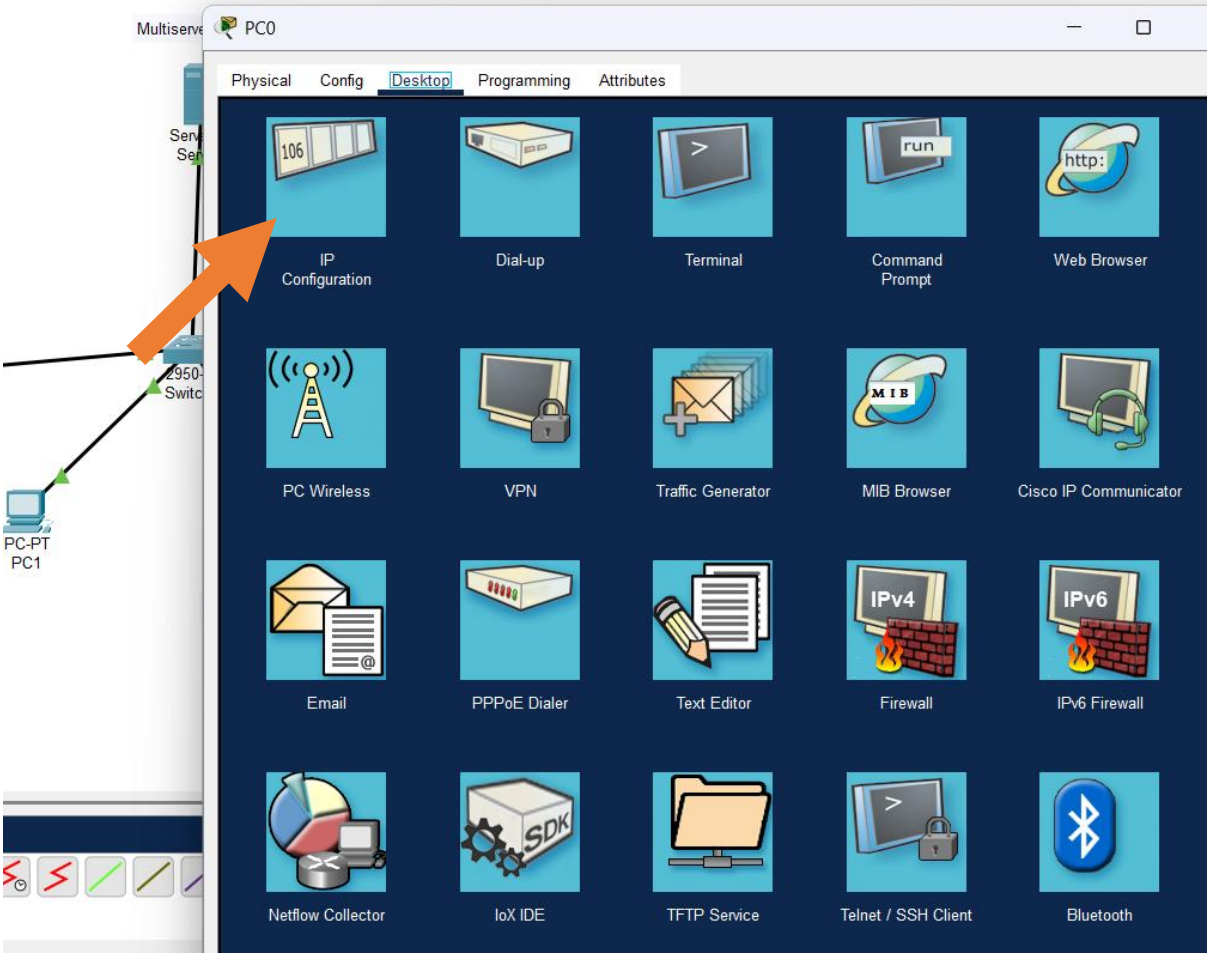


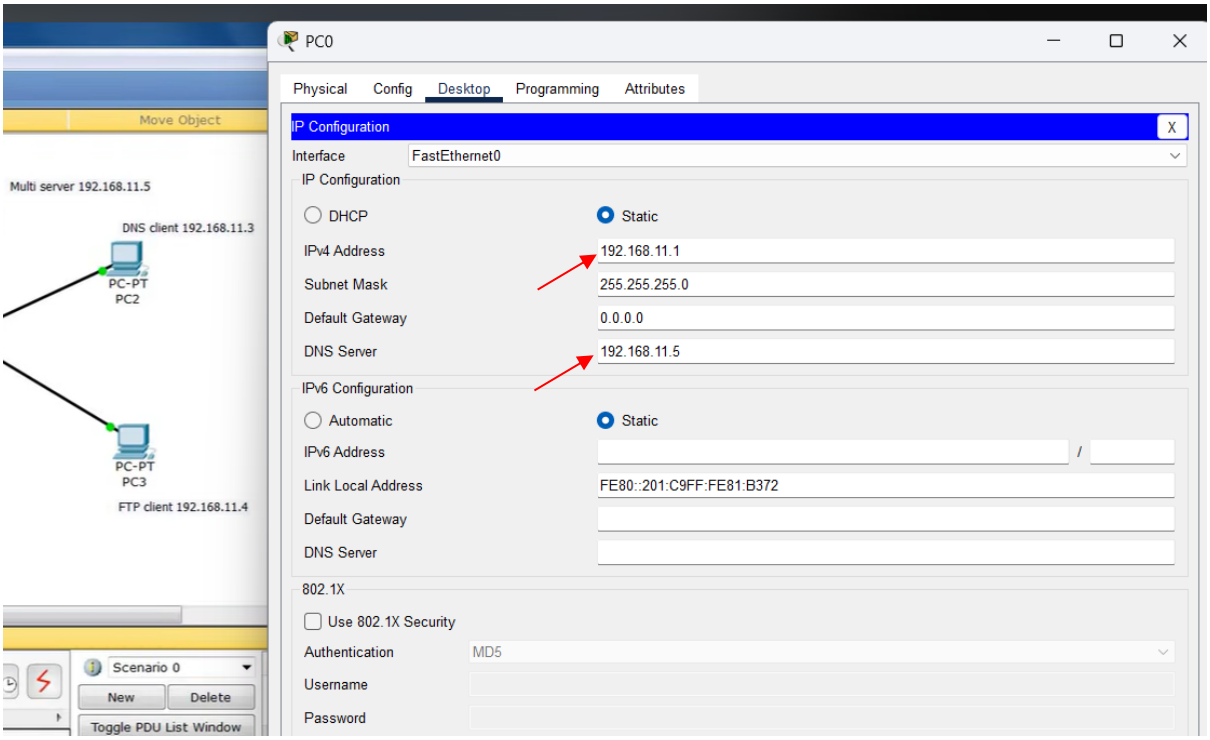

2. Name all Nodes



## 3. Configure IP address for each node
## Click on PC0 and follow given steps



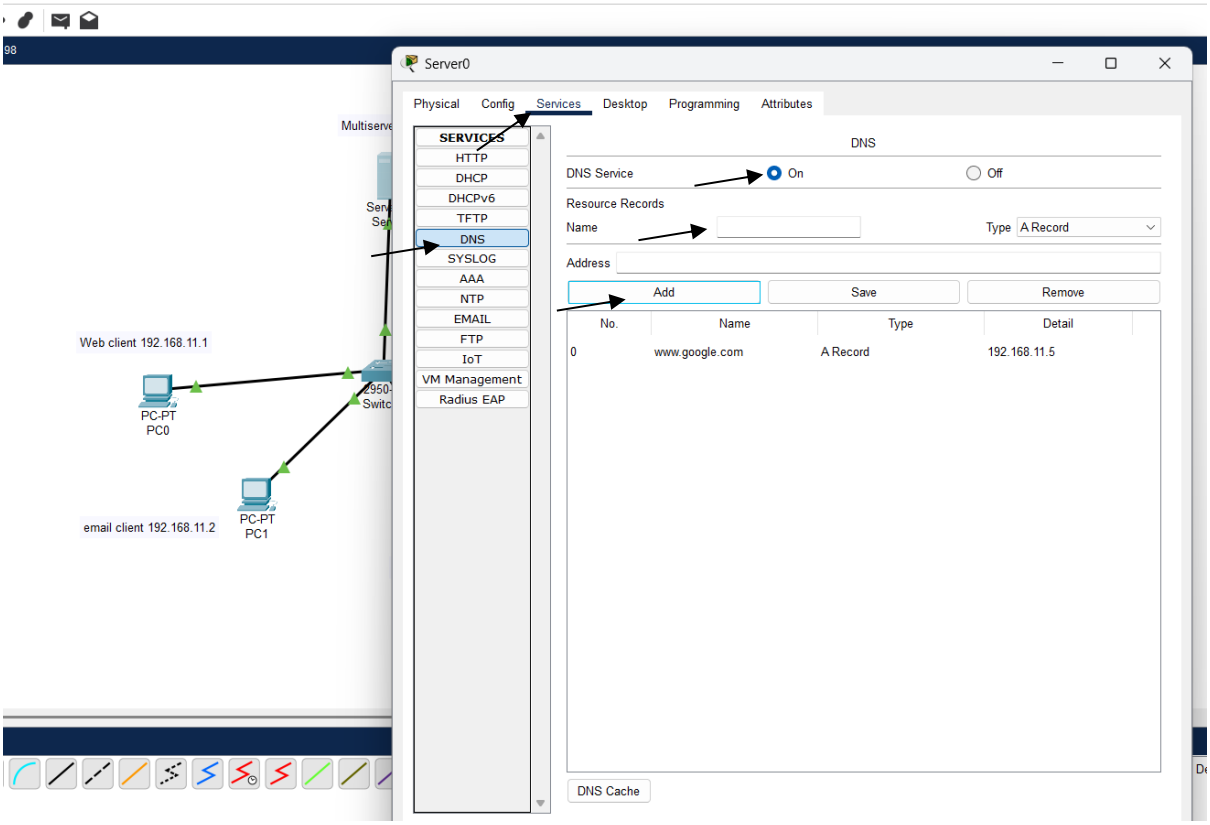





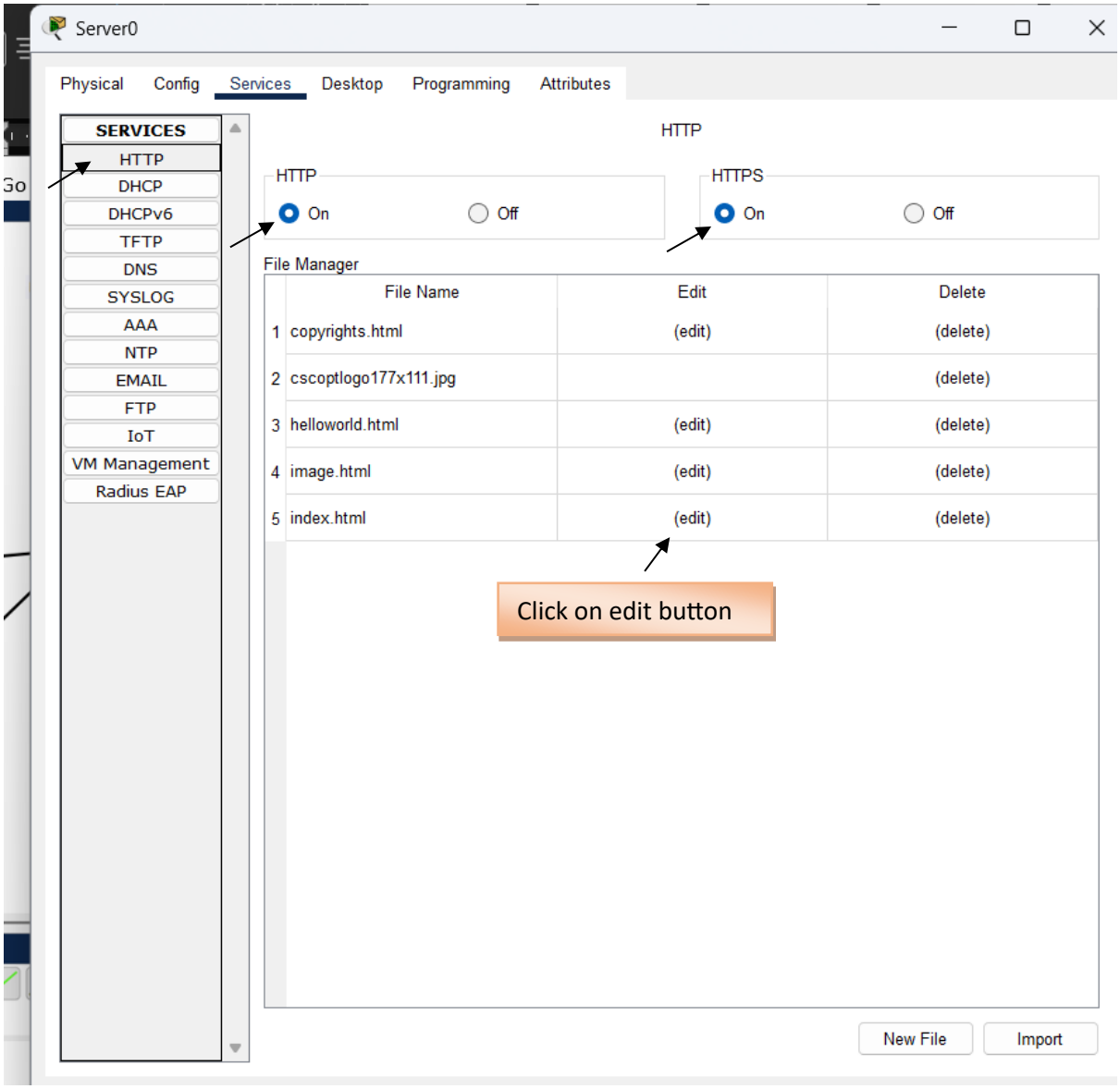

- **Repeat this step for all remaining nodes**

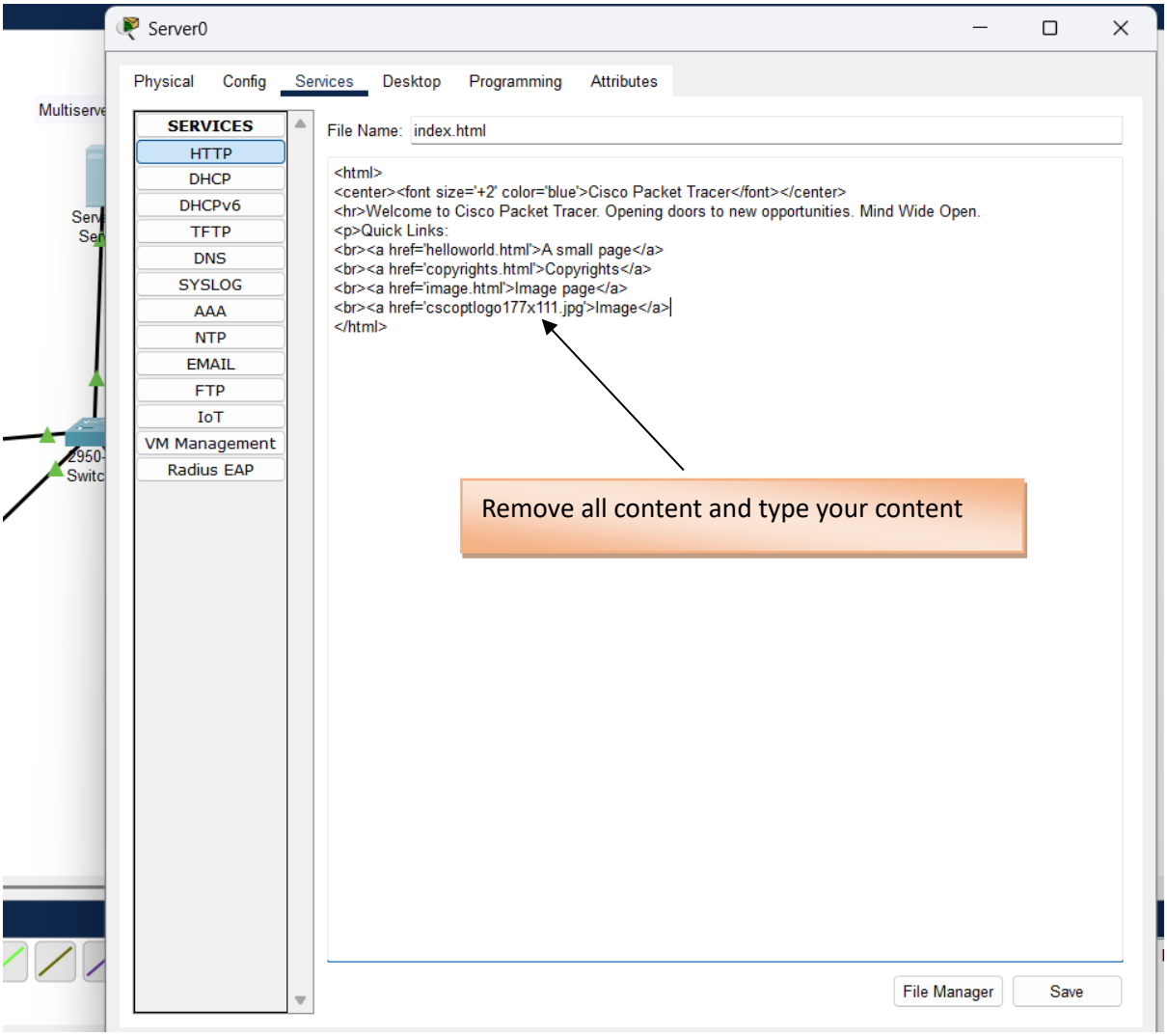
4. **Click on server assign IP address as given in above step**

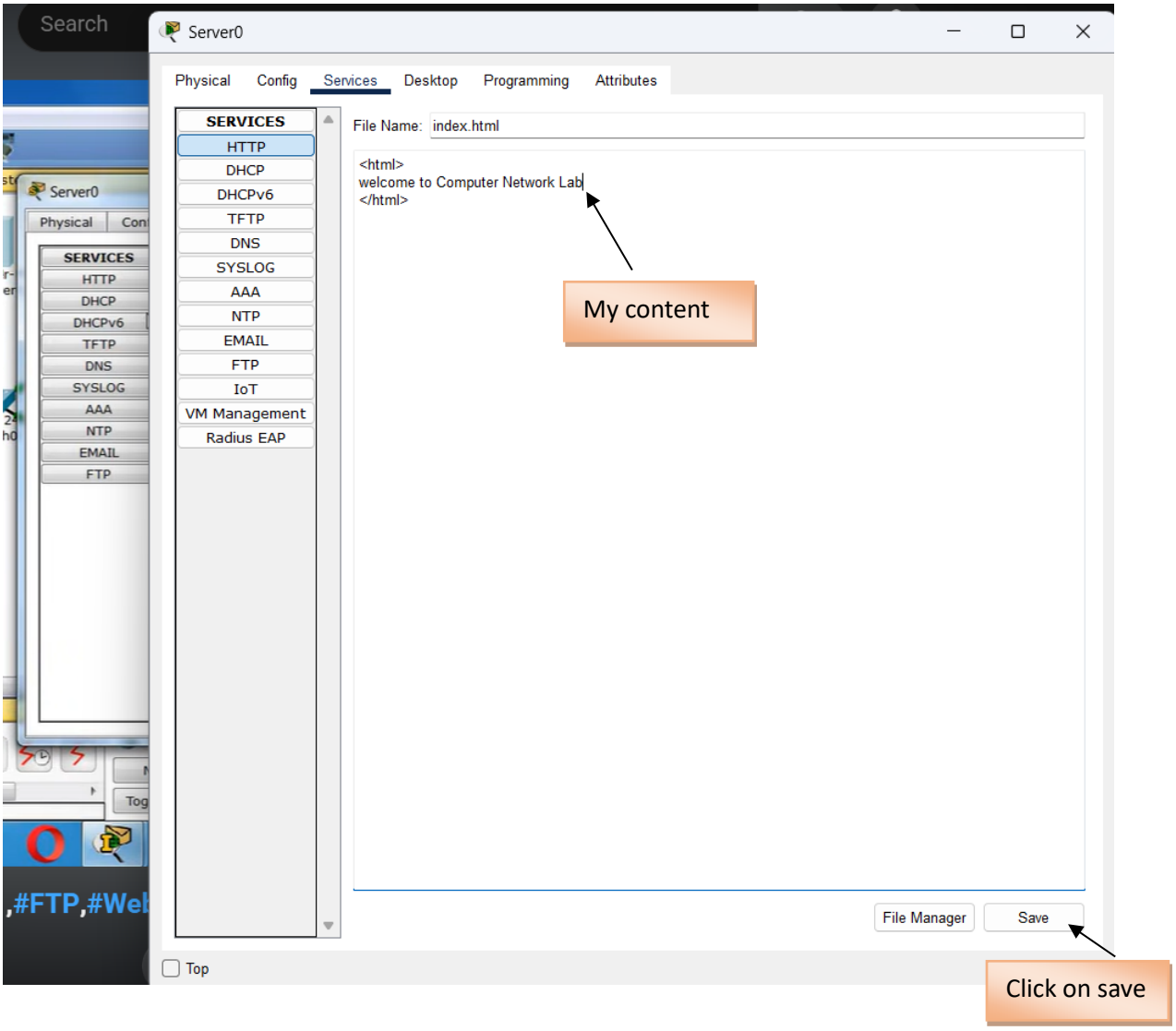
5. **Again click on server go to services ⟶ DNS⟶ Turn DNS service ON Give Name – [www.google.com](http://www.google.com) and click on add button.**



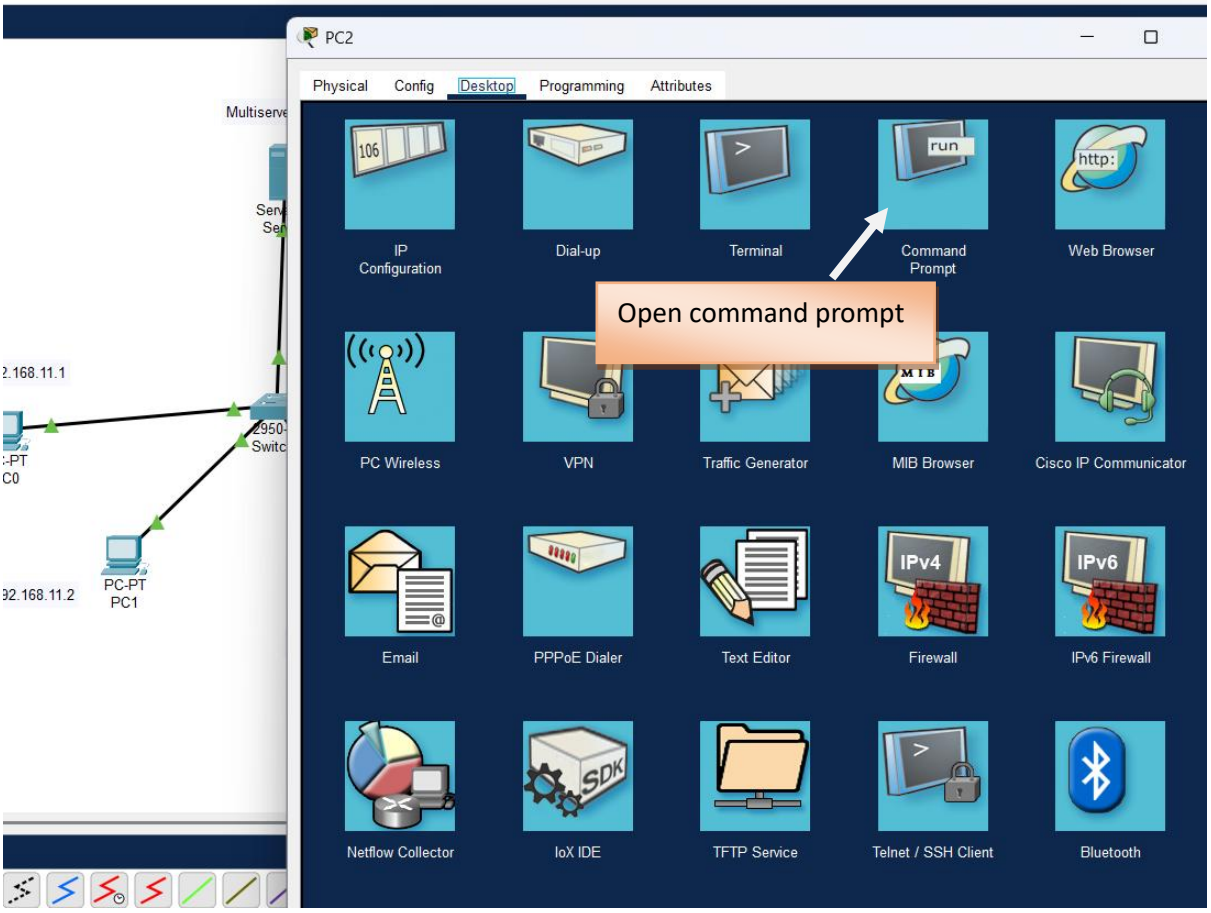
6. Go to HTTP server

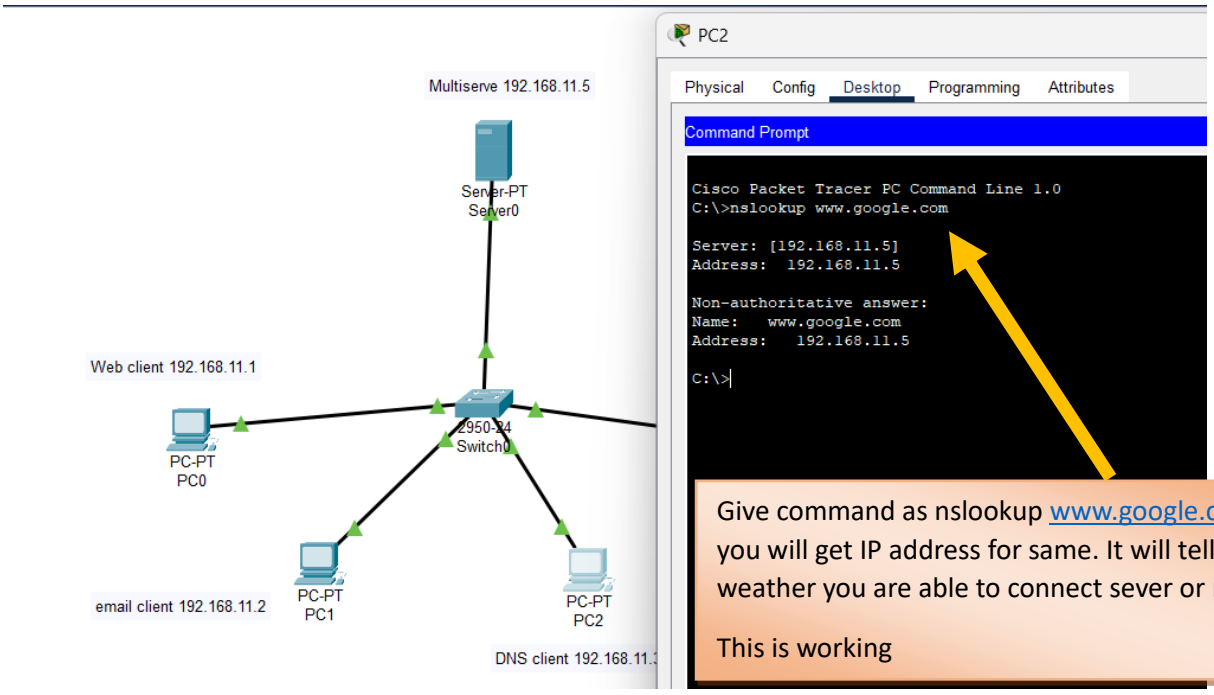



Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
File Name: index.html
<html>
<font size='+2' color='blue'>Cisco Packet Tracer</font>
<hr>Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
<p>Quick Links:
<br><a href='helloworld.html'>A small page</a>
<br><a href='copyrights.html'>Copyrights</a>
<br><a href='image.html'>Image page</a>
<br><a href='cscoptlogo177x111.jpg'>Image</a>
</html>
Remove all content and type your content
File Manager
Save




7. **Click on DNS PC2**



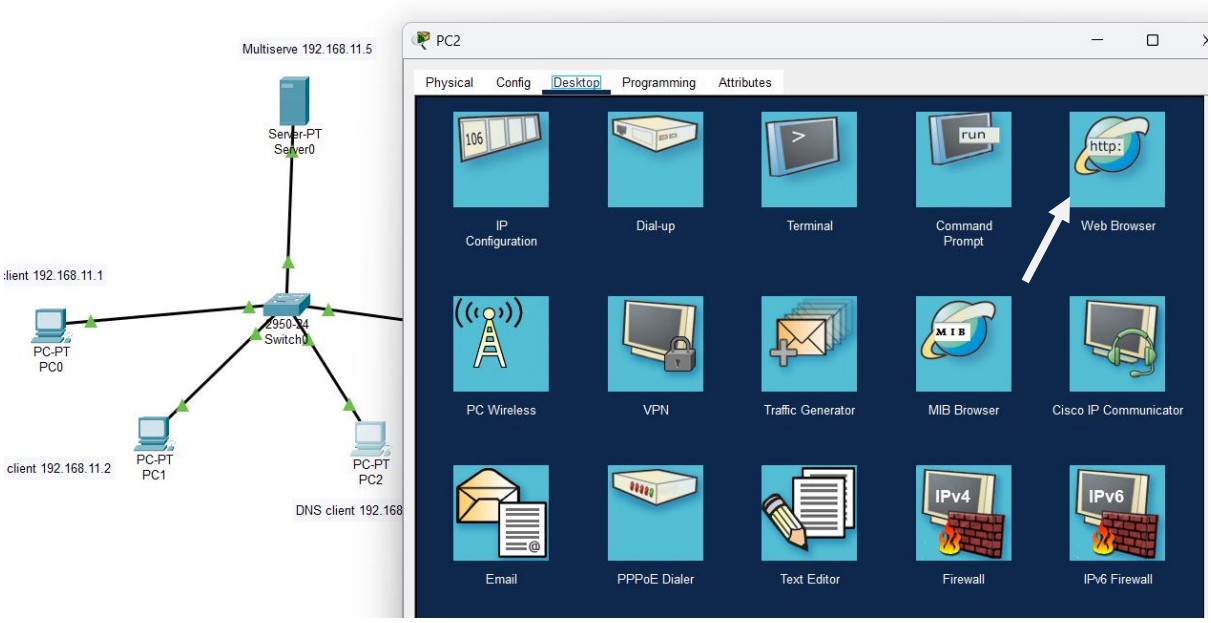



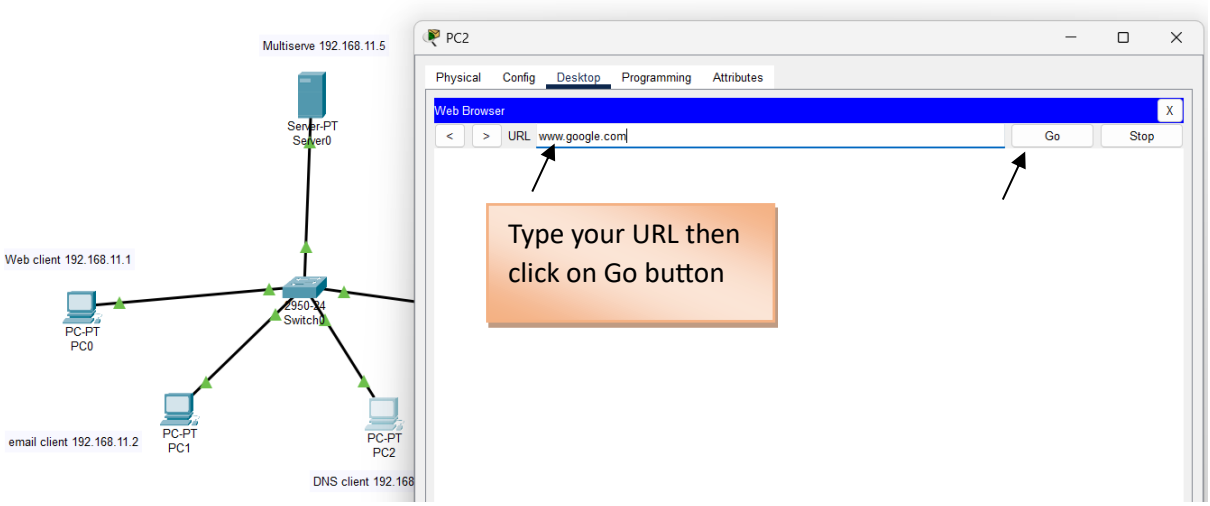

**When nslookup is executed, the client sends a DNS request to the DNS server.**
**The DNS server replies with the corresponding IP address.**
**DNS communication uses UDP (port 53) because it is fast and requires only a request and a response.**
**In Simulation Mode, you can filter only DNS packets and observe the UDP packets travelling between the client and the server.**

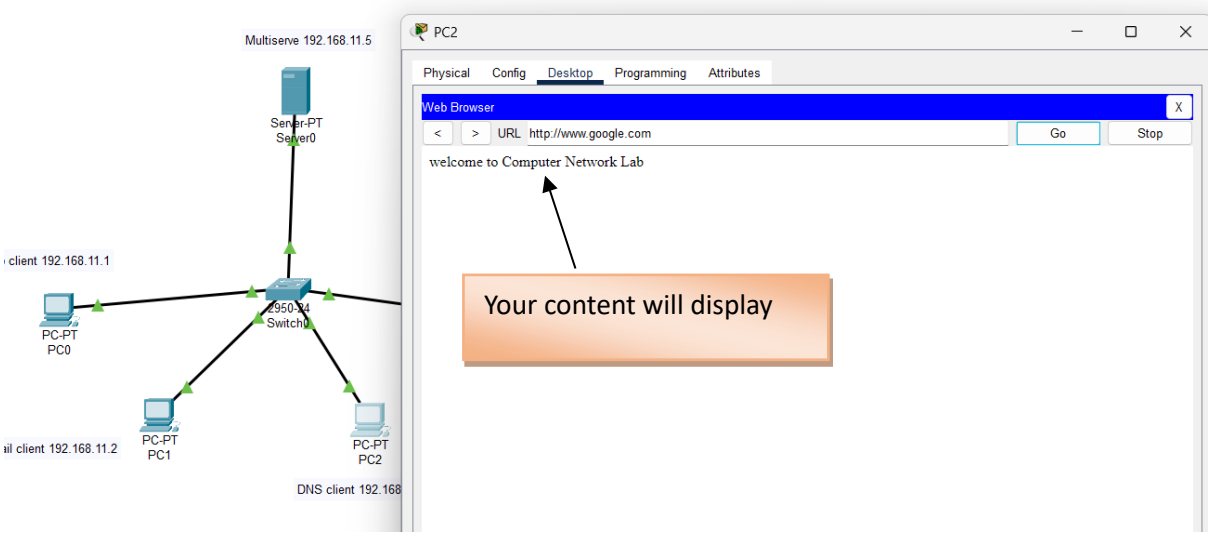
- **Every network service communicates through a port number.**
- **Port numbers identify the application or service running on a computer.**
- **DNS servers listen on port 53.**
- **When a client executes:**
  **nslookup www.google.com**
  **it sends a DNS query to UDP port 53 on the DNS server.**
- **The server processes the request and sends back the corresponding IP address.**
  **Common Port Numbers**

| Service | Protocol | Port Num |
|---|---|---|
| DNS | UDP (mostly), TCP (zone transfer/large responses) | 53 |
| HTTP | TCP | 80 |
| HTTPS | TCP | 443 |
| SMTP | TCP | 25 |
| POP3 | TCP | 110 |
| FTP | TCP | 21 |

**8. Go on web browser**






**TCP Traffic Generation (HTTP)**

**Step 1: Configure the Network**

- **Assign IP addresses to the client and server.**
- **Configure the DNS server.**
- **Enable the HTTP service.**
- **Create a web page on the server.**

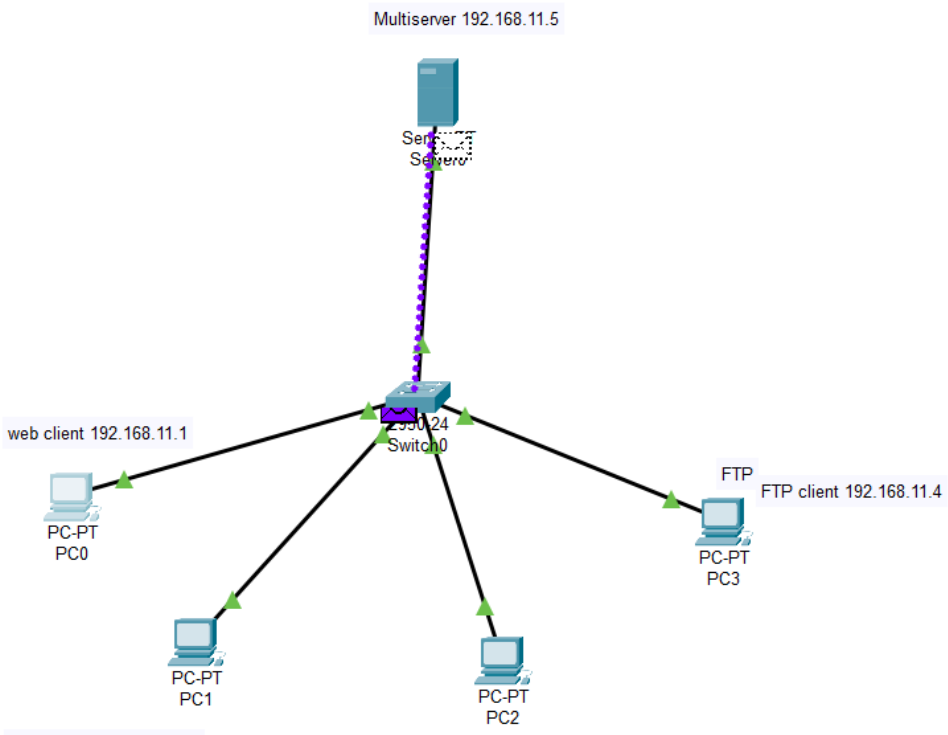
**These steps are already performed**

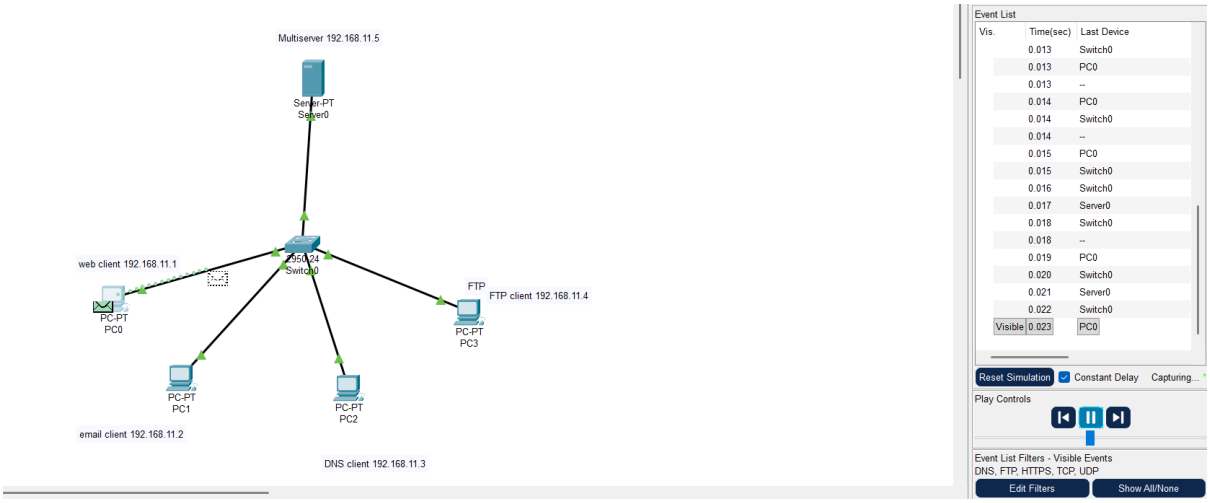
**Step 2: Switch to Simulation Mode**

- **Click Simulation at the bottom-right corner of Packet Tracer.**
- **Keep TCP, HTTP, and DNS protocols enabled in the Event List Filters.**

**Step 3: Open the Web Browser**
**On PC0:**
**Desktop → Web Browser**
**Type:**
**www.google.com**
**Click Go.**

**Start simulation**



**Step 4: DNS Resolution (UDP)**
**Before TCP starts, the client must know the server's IP address.**

```
Client                DNS Server
  |----DNS Request------->|
  |<---DNS Reply----------|
```

**The client now knows the server's IP address.**

---

**Step 5: TCP Three-Way Handshake**
**Now TCP establishes the connection.**

```
Client                      Server
  |                           |
  |---------- SYN ----------->|
  |                           |
  |<------- SYN + ACK --------|
  |                           |
  |---------- ACK ----------->|
  |                           |
```

**Explain each packet:**
**SYN**

- **Sent by the client.**
- **Means: "I want to establish a connection."**

**↓**
**SYN-ACK**

- **Sent by the server.**
- **Means: "I received your request and I'm ready."**

**↓**
**ACK**

- **Sent by the client.**
- **Means: "Connection established."**

**After this step, the TCP connection is established.**

---

**Step 6: HTTP Data Transfer**

```
Client                      Server
  |                           |
  |------ HTTP GET Request ---------->|
  |                           |
  |<----- HTTP Response (Web Page) ---|
  |                           |
```

**The client requests the webpage, and the server sends it back reliably using TCP.**

---

**Step 7: TCP Connection Termination**
**After the webpage is displayed, TCP closes the connection.**

```
Client                      Server
  |                           |
  |---------- FIN ----------->|
```

```
  |                          |
  |<---------- ACK ----------|
  |                          |
  |<---------- FIN ----------|
  |                          |
  |---------- ACK ---------->|
```

**The TCP session is now closed.**

---

**Complete Flow Diagram**

```
Client                       Server
  |                          |
  |---- DNS Request (UDP) ---------->|
  |<--- DNS Reply (UDP) ----------|
  |                          |
  |---- SYN ---------------------------->|
  |<--- SYN + ACK ----------------------|
  |---- ACK ---------------------------->|
  |                          |
  |---- HTTP GET ------------------------>|
  |<--- HTTP Response ----------------------|
  |                          |
  |---- FIN ---------------------------->|
  |<--- ACK ----------------------------|
  |<--- FIN ----------------------------|
  |---- ACK ---------------------------->|
```

---

**Explain While Running the Simulation**

1. **Click Capture/Forward.**
2. **First, students will see the DNS packets (UDP).**
3. **After DNS completes, they will see SYN, SYN-ACK, and ACK packets. Explain that this is the TCP three-way handshake.**
4. **Next, they will see the HTTP GET Request and HTTP Response packets.**
5. **Finally, they will observe the FIN and ACK packets used to terminate the TCP connection. This sequence clearly demonstrates how TCP establishes a reliable connection, transfers data, and then closes the connection.**